

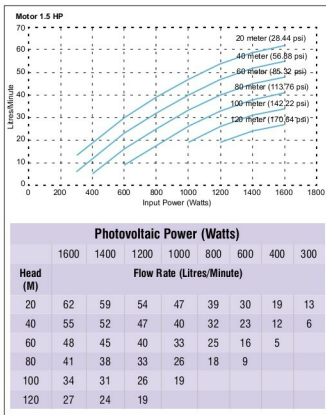


Fig. 3: 1.5HP motor pump performance

TABLE III
Capacities of DC Surface Pump and AC Submersible Pump

Type	Solar panel pump capacity	Head	Discharge rate/day (in litres)
DC Surface	900Wp 1HP 1800WP 2HP	14m 14m	75,000 140,000
AC Submersible	1800WP 0.75HP	50m	50,000

ing. "One should buy AC submersible pump, because the maintenance for submersible is almost nil as compared to DC pump," says Hitesh Kapuriya, assistant manager, JJ PV Solar Pvt Ltd.

Indian manufacturers of pump-sets for various applications can be found at <http://www.pumpsindia.com> website. The type of motor pump, the wattage of the solar module, accessories, etc are some of the decisions you have to make. So before buying the system you should check the following:

1. *Solar water pump installation site.* You should first check whether it is to be installed near a pond, river, well or

somewhere else.

2. *Pump setting depth.* The accessories required for underground installation would be different and add to overall cost.

3. *Requirement of daily water.* If you need more water per day, you need heavier and powerful equipment.

4. *AC or DC pump.* For AC pump, you need solar panel, pump controller (inverter), module mounting assembly and accessories like cable, connectors and pipe. For DC pump, pump controller would not be required in case of direct connection with PV panel.

In addition, you may consider installing solar trackers and pump protection units. The com-

plete system should be tested and certified by an authorised test centre to get the best performance and water discharge norms as specified by the ministry.

Various capacities of DC surface and AC submersible pumps available from a vendor are listed in Table III, just for information.

Current scenario in India

Nearly sixty per cent of Indians work as farmers in the field of agriculture. But due to non-availability of electricity in remote areas, irrigation becomes difficult even if the land is fertile as it

depends majorly on rainfall only. Solar energy can save not only electricity but also save Indian farmers (during drought and scanty rainfall) who are otherwise prone to suicide. Solar water pumping systems can make irrigation possible even in remote areas. The pumps are available for various types of irrigation, including paddy fields, horticulture farms, gardens, etc.

"The present scope of solar irrigation system in India is very good because there is support from nodal agencies in the states like Rajasthan, Bihar, etc. The Ministry of New and Renewable Energy, government of India, gives 30 per cent subsidy for a five horsepower solar water pump set," says Hitesh. Unfortunately, "Average cost of solar water pumping system is ₹ 450,000. Government policy for subsidy is having restriction up to 5HP solar pump only," adds Hitesh.

Besides the high cost of solar irrigation systems at present, another problem is the lack of awareness of the benefits amongst the farmers and consumers.

The future scope

With advancements in technology, the systems can be made more user-friendly. You can control the motor pump from a remote location using your mobile phone. You can get water level indication of the reservoir and overhead storage tank in your mobile phone. These technologies are already available in the market, though presently they are not so popular.

For further information on this subject you may refer to 'Irrigation Automation Opportunities in Rural Areas' article published in January 2012 issue. Also, you may refer to DIY articles on related topics, such as 'Cellphone-Based Remote Controller for Water Pump' published in January 2013 and 'GSM-Based Borewell Water-Level Monitor' published in May 2012 issues. ●

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